

# Towfiqul Islam

Senior Software Engineer | Java / Spring Boot & PHP / Laravel | Dhaka, Bangladesh

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Portfolio: <https://towfiq-ul.github.io>

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## Professional Summary

Senior Software Engineer with 7+ years designing and shipping backend systems across two stacks: Java / Spring Boot microservices for high-throughput fintech platforms, and PHP / Laravel applications delivered for government, private company, and open-source engagements. Most recently built core payment platform components as Augmented Staff at bKash Limited, Bangladesh's #1 Unicorn MFS. Equally comfortable owning a Spring Boot microservice with Kafka and DynamoDB, or a Laravel codebase end-to-end — schema design, Eloquent, queues, Blade views, and RBAC. Passionate about clean architecture, system reliability, and shipping things that scale, in either stack.

## Core Expertise

- **Backend:** Java (11–21), Spring Boot, Spring MVC, Spring Security, Spring Integration, Spring Cloud, PHP, Laravel 9–12, Composer, Eloquent, Blade, Microservices, REST APIs, gRPC
- **Frontend Integration:** React, TypeScript, Vite, jQuery
- **Cloud & DevOps:** AWS (EC2, S3, SNS, SQS), Docker, Kubernetes, Jenkins, GitHub Actions CI, CI/CD
- **Databases:** MySQL, Oracle, PostgreSQL, DynamoDB, Cassandra, Redis, JPA / Hibernate
- **Messaging:** Kafka, RabbitMQ, ActiveMQ, AWS SNS/SQS
- **Testing:** JUnit, Mockito, TestNG, Rest Assured, Postman
- **Security & Auth:** OAuth2, JWT, KeyCloak, LDAP
- **Practices:** Agile/Scrum, Domain-Driven Design, Event-Driven Design, clean architecture, code review, mentoring

## Work Experience

### Senior Software Engineer – Exabyting

*April 2024 – August 2026 | Dhaka, Bangladesh*

Augmented resource at bKash Limited. Built and maintained core backend services (Customer App MW, FinRec, PIN Reset MW) on Java / Spring Boot, DynamoDB, and Kafka, processing 2.5M+ transactions daily. Led the Spring Boot 2→3 migration on the PIN Reset service (javax→jakarta, AWS SDK v1→v2, Jedis→Lettuce) for zero-downtime deployment. Mentored junior engineers on best practices, release readiness, and system architecture.

### Software Engineer – SELISE

*November 2021 – April 2024 | Zurich, Switzerland (Remote)*

Extended ThingsBoard (Java, open-source IoT platform, merged contributor) into a multi-tenant IoT middleware platform for a Swiss enterprise customer, modeling and optimizing time-series telemetry at a scale of millions of data points.

### Software Engineer – Exabyting

*December 2019 – November 2021 | Dhaka, Bangladesh*

Developed both Java and PHP/Laravel government and private web applications and services, including PREVENTO (a Laravel healthcare app backend) and internal Java tooling for bKash Limited. Worked as partner team with bKash Limited's Solution Engineering Team.

## Key Projects

- **bKash App** – Consumer mobile financial app; Java 11–21 / Spring Boot (WebFlux) microservices, DynamoDB, Kafka, AWS, gRPC, Kubernetes
- **Aerotia International** – Laravel 12 (PHP 8.2) JSON API + Blade admin CMS, React 18/TypeScript frontend
- **Tesenso MW** – Multi-tenant IoT platform (extended ThingsBoard) for an EV/aerospace client
- **bKash-FinRec** – Financial reconciliation system with LDAP auth, custom AOP access control, SIEM audit logging
- **PIN Reset MW** – Spring Boot SOAP+REST middleware bridging Huawei CPS, JAXB codegen from WSDL/XSD

- **AMS** – Airtime Management System; automated CSV/XLS/PDF generation, SFTP operations, maker-checker approval flow, TPS-limiting service with interval-based retry
- **Aerotia Accounting** – Laravel 9 ledger/accounting admin portal, Blade, Docker
- **HRMS User & Role Management** – Modular Laravel RBAC package
- **laravel-gitstamp** – Open-source Composer package, MIT-licensed, CI-tested across Laravel 10–12
- **PREVENTO** – Laravel backend for a healthcare Android app (Google Play), doctor-patient messaging, Firebase-integrated

## Key Impacts

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- **bKash App**: built and maintained services processing 2.5M+ transactions/day
- **AMS**: reduced HTTP 429 rate-limit exceptions by solo-designing and building a self-contained, single-responsibility TPS-limiting service with interval-based retry, later made dynamically configurable and scalable to support Airtime Mobile Recharge Automation
- **Spring Boot 2→3 migration**: led the migration on the PIN Reset service (javax→jakarta, AWS SDK v1→v2, Jedis→Lettuce) for zero-downtime deployment
- **Reactive Kafka**: implemented reactive Kafka integration using spring-kafka and reactor-kafka to improve throughput and resilience across core banking microservices
- **Tesenso MW**: extended to handle millions of IoT data points across a multi-tenant platform for an EV/aerospace client
- **API latency**: reduced latency by replacing REST with GraphQL for client-facing data-fetching endpoints, eliminating over-fetching and cutting round trips for nested resources
- **CI/CD**: cut build time from 8m 29s to 4m 56s (~42% faster) by implementing multithreaded build execution in the pipeline
- **Developer tooling**: built a Bash/shell pipeline streaming live logs from an AWS EC2 instance to developers' local machines, cutting log-check time by ~15 minutes per check

## Education

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B.Sc. in Computer Science & Engineering – International Islamic University Chittagong (2014 - 2019)

## Certifications & Awards

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- Java (Intermediate), C++ (Advanced), Python (Basic) – HackerRank
- IIUC IUPC 2017 – 2nd Position (Coach)
- EDU Programming Contest 2018 – 5th Position (Coach)
- CSE Week Programming Contest 2016 – 5th Position
- Solved 500+ competitive programming problems

## Open Source Contributions

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- ThingsBoard (Java, open-source IoT platform) – Pull Request #6614 (Merged)
- laravel-gitstamp (PHP/Laravel) – Author & Maintainer, MIT-licensed, CI-tested across Laravel 10–12
- chatling – Framework-agnostic AI chat widget (npm), powers this portfolio's own AI assistant
- jmbus (Java) – Author & Maintainer; JNA wrapper around the M-Bus (Meter-Bus) protocol library libmbus, grown out of Tesenso IoT platform work
- claude-auto-resume-command – Author & Maintainer; Claude Code /auto-resume slash command plus a Stop-hook-based session guardian that auto-resumes paused sessions